Year 12 Maths - Similar to June 2023 Question 2

Similar questions - Pure & Mechanics

Two new questions on the same topic as the exam question, a little more open-ended. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

The diagonal of a square has length $(4+2\sqrt{2})$ cm.

(a) Show that the side of the square is $(2+2\sqrt{2})$ cm. **[3]**

(b) Find the area of the square in the form $a+b\sqrt{2}$, where a and b are integers. **[2]**

2

(a) Show that $2^{n+3}-2^n = 7\times 2^n$. **[2]**

(b) Given that $2^{n+3}-2^n = 224$, find the value of n. **[2]**

(c) Show that $3^{n+2}-3^n$ is a multiple of 8 for every positive integer n. **[2]**

Question 1

(a) Side $= \dfrac{4+2\sqrt{2}}{\sqrt{2}} = \dfrac{(4+2\sqrt{2})\sqrt{2}}{2} = \dfrac{4\sqrt{2}+4}{2} = 2+2\sqrt{2}$

(b) Area $= (2+2\sqrt{2})^2 = 4+8\sqrt{2}+8 = 12+8\sqrt{2}$ cm^2

Question 2

(a) $2^{n+3}-2^n = 2^n(2^3-1) = 7\times 2^n$

(b) $7\times 2^n = 224 \Rightarrow 2^n = 32 \Rightarrow n = 5$

(c) $3^{n+2}-3^n = 3^n(9-1) = 8\times 3^n$, which is a multiple of 8